SIMATS ENGINEERING

Department of Information Technology

ITA0507 – COMPUTER VISION

ASSIGNMENT 1 – CO1

Apply, Compare and Justify Image Acquisition, Formation and Processing Techniques for a Vision-Based E-Waste Battery Sorting Line

CO1: Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems.

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SCENARIO – VISION-BASED E-WASTE BATTERY SORTING LINE

GreenCycle Robotics Corp. operates an e-waste reclamation facility. Mixed consumer batteries arrive on a bulk conveyor and are currently sorted manually. Operators experience eye fatigue, miss dangerous swelling and struggle to

read small laser-etched chemistry codes under changing factory illumination. Incorrectly mixing lithium cells with nickel-metal hydride (NiMH) batteries can create a serious downstream safety hazard.

The proposed computer vision system is installed above a single-lane conveyor. It must classify cylindrical 18650 cells, cylindrical 21700 cells and prismatic pouch cells; read chemistry labels such as LCO, LFP, NMC and NiMH; detect rust, dents, cracks and pouch swelling; and activate a pneumatic kick-rejecter.

Parameter	Given operating value	Engineering significance
Conveyor speed	1.2 m/s	Determines motion blur and frame timing
Field of view	320 mm × 240 mm	Sets spatial sampling requirement
Target objects	10–70 mm length; up to 30 mm height	Requires multiple form-factor classes
Throughput	15 cells/s	One cell arrives every 66.7 ms on average
Decision deadline	400 ms; piston 0.48 m downstream	Available travel time is 0.48/1.2 = 0.4 s
Smallest defect	0.3 mm micro-crack / rust pit	Requires high spatial sampling and suitable optics
Etched text stroke	0.4 mm	Requires at least two samples across stroke as a basic Nyquist condition
Swelling precision	±0.5 mm	Requires calibrated geometry and controlled pose
Rust contrast	5–8 grey levels from healthy casing (8-bit)	Requires stable illumination and adequate bit depth
Ambient illumination	1,500–45,000 lux; 100 Hz flicker	Requires enclosure / strobed lighting
Camera height	400 mm above belt	Used for focal length and perspective calculations
Compute	4 CPU cores, 16 GB RAM, no GPU	Favours efficient classical CV and compact models
Archive	1.5 TB SSD; retain every cell image for 5 days	Requires compression, event-based ROI and retention planning
Budget	₹1,25,000	Constrains sensor, lens, lighting and controller choice
Maximum error rate	≤1.5%	Must be demonstrated on representative validation data

COMMON ASSUMPTIONS AND EXPERIMENTAL METHODOLOGY

The numerical work below is an engineering design analysis based on the operating data in the assignment brief. Values labelled calculated are derived directly from the supplied parameters. Values labelled design target or estimate are proposed engineering targets, not claimed experimental measurements. A real commissioning report must replace estimates with measured results from a labelled battery dataset and the actual installed camera.

Item	Assumption used	Reason
Image rate	15 frames/s for one image per cell	Matches the maximum cell throughput for firstorder bandwidth calculations
Colour image storage	3 bytes/pixel for uncompressed 8-bit BGR	Standard 8-bit, three-channel representation
Sensor format	1/2-inch sensor with approximately 6.4 mm active width	Representative active width for focal-length estimate; exact sensor datasheet must be checked
Optical scale	Horizontal FOV is 320 mm at 400 mm standoff	Given camera geometry
Storage duration	5 × 24 × 60 × 60 = 432,000 seconds	Continuous five-day archive calculation
GigE practical payload	Below 125 MB/s theoretical byte rate	Protocol overhead and packet gaps reduce usable throughput
Accuracy claims	No unexecuted model is assigned measured accuracy	Prevents unsupported performance claims

Overall system objective

The system should produce a traceable record containing the camera frame, cell identifier, predicted form factor, OCR chemistry string and confidence, defect measurements, final accept/reject decision, timestamp and actuator command.

Any uncertain chemistry read or safety-defect result should be routed to a reject or manual-review path rather than silently accepted.

TASK 1 – SYSTEMS COMPARISON: MANUAL, ASSISTED AND AUTOMATED SORTING [4 MARKS]

The first task compares three operating workflows. Technique A depends on human visual inspection. Technique B uses image processing to improve the image but leaves the final decision to the operator. Technique C performs the image analysis and decision automatically, then controls the pneumatic rejecter.

1.1 Implementation logic of the three techniques

Technique	Processing sequence	Strength	Main limitation
A – Manual	Battery on belt → operator observes casing and code → operator sorts	Human can use context and recognize unfamiliar defects	Fatigue, inconsistent judgement, handling risk and limited speed
B – Image-processing-assisted	Camera → brightness/contrast enhancement → digital zoom → operator display → operator decision	Improves visibility of faint rust and small text without full automation	Human decision time remains; screen delay and fatigue remain
C – Automated computer vision	Camera → calibration → segmentation → features/OCR/classifier → safety rules → PLC → piston	Repeatable, non-contact, measurable and traceable	Needs representative training/validation data and safe fallback logic

1.2 Throughput and timing analysis

The line handles 15 cells each second. Therefore, the average interval between consecutive cells is:

Cell interval = 1 / 15 = 0.0667 s = 66.7 ms.

A human operator generally needs visual inspection time for orientation, reading the chemistry code and checking surface defects. Even if an operator takes only 1.0 second per cell, the rate is approximately 1 cell/s. The number of parallel operators required is approximately 15 operators at 1 second per cell, before allowing for breaks, overlap and handling delays. If inspection takes 1.5 seconds per cell, the theoretical requirement becomes 22.5 operators.

Measure	A – Manual	B – Assisted	C – Automated design target
Processing capacity	~0.7–1.0 cell/s per operator (planning estimate)	Human-limited; image display does not remove decision time	15 cells/s target; pipeline must process each arrival
Repeatability	Variable; operator-dependent	Improved viewing, but final decision remains variable	Deterministic for identical inputs and fixed software
Objectivity	Low to medium	Medium	High when calibration and rules are controlled
Non-contact	May involve manual handling	Camera viewing is non-contact	Fully non-contact inspection and rejection
Continuous operation	Limited by fatigue and shift length	Still limited by operator endurance	Can operate continuously with scheduled maintenance
Traceability	Paper or manual log	Saved images possible	Image + measurements + OCR + decision + timestamp
New defect type	Human can adapt through instruction	Human can learn from enhanced display	Requires rule update or retraining and validation
Lifecycle cost	Recurring labour cost	Labour plus camera/display cost	Higher initial integration cost; lower per-cell labour

1.3 Agreement and Cohen's kappa

For a pilot study, each operator should inspect the same labelled battery set twice. Inter-rater agreement compares different operators; intra-rater agreement compares the same operator at two different times. Cohen's kappa removes agreement expected by chance:

$\kappa = (p_o - p_e) / (1 - p_e)$, where p_o is observed agreement and p_e is chance-expected agreement.

A kappa near 1 indicates very strong agreement; a value near 0 indicates agreement close to chance. The assignment brief does not provide real operator decisions, so no measured kappa value can be honestly reported here. The required

experiment is to record a binary decision (accept/reject) for each sample, calculate pairwise kappa for each operator pair, and report the mean and confidence interval.

Validation output	How it is obtained	Acceptance interpretation
Inter-rater κ	Pairwise Cohen's kappa across operators on the same images	High κ indicates consistent sorting decisions
Intra-rater κ	Same operator repeats the set after a time gap	Low change indicates repeatability
False accept rate	Unsafe cells accepted / all unsafe cells	Critical safety metric; should be minimized
False reject rate	Good cells rejected / all good cells	Measures avoidable product loss
Latency	Capture timestamp to PLC decision timestamp	Must remain below the 400 ms travel deadline

1.4 Image processing, computer vision, machine vision and computer graphics

Field	Input → output	Role in the battery line
Image processing	Image → enhanced / filtered image	Denoising, contrast enhancement, flat-field correction and resizing
Computer vision	Image → interpreted information	Finds cell type, OCR text, defects and dimensions
Machine vision	Sensor + optics + lighting + software + actuator	Complete industrial inspection and sorting system
Computer graphics	Model / scene description → rendered image	Can create synthetic training images and visual simulations; it does not inspect the real battery

1.5 Recommended approach

Technique C is the appropriate production direction because the line rate is 15 cells/s and the task includes safetycritical chemistry separation. Technique B is useful during commissioning and for operator review, while manual inspection should be retained as an exception-handling process. Automation must not accept a cell when OCR confidence is low, the chemistry code is unknown, or a defect measurement lies near the decision threshold. Such cases should be diverted to a safe reject or review lane.

Decision: Select automated machine vision with a human-review fallback. This recommendation is conditional on proving the required ≤1.5% error rate on a representative, independently labelled validation set.

TASK 2 – SYSTEM ARCHITECTURE: PIPELINE LEVEL DEPTH COMPARISON [5 MARKS]

A computer vision system can be organized into low-level, mid-level and high-level processing. Each additional level converts raw pixels into more meaningful information, but it also adds computational work and design complexity.

2.1 Three pipeline depths

Technique	Operations	Decision mechanism	Expected behaviour
A – Low-level only	Gaussian blur, adaptive threshold, pixel counts and binary masks	Fixed threshold on dark / bright pixel count	Fast, but shadow boundaries and oil stains can trigger false alarms
B – Low + mid-level	Contours, bounding rectangles, moments, aspect ratio, circularity and connected components	Expert rules based on geometry and surface regions	More robust to noise; touching objects may merge into one contour
C – Low + mid + high-level	Low/mid features passed to SVM or k-NN classifier; OCR and confidence checks	Classical ML decision plus explicit safety rules	Can combine multiple features, but depends on representative labelled data

2.2 Low-level processing

Low-level processing operates directly on pixel intensities. A Gaussian filter reduces high-frequency noise. Adaptive thresholding calculates a local threshold and creates a binary mask. A simple rule can count pixels that are darker than a calibrated reference. This is computationally efficient, but it does not understand whether a dark region is rust, a shadow, an oil mark or a printed label.

A shadow boundary can create a large connected dark region. If the algorithm uses a fixed dark-pixel count, the shadow may exceed the defect threshold even when the battery surface is healthy. Adaptive thresholding reduces some illumination effects but can also erase low-contrast rust or split one defect into multiple fragments.

2.3 Mid-level geometrical features

Mid-level processing extracts objects and regions. Contours provide boundaries; cv2.minAreaRect estimates a rotated bounding rectangle; aspect ratio helps separate long cylindrical cells from compact pouch cells; contour circularity helps identify approximately circular end views. Moments provide area and centroid. These features are more meaningful than raw pixel counts.

Feature	Formula / meaning	Use
Area	Number of foreground pixels × pixel area	Approximate projected size
Aspect ratio	Long side / short side of bounding rectangle	Form-factor separation
Circularity	$4\pi A / P^2$, where A is area and P is perimeter	Distinguishes round silhouettes from irregular masks
Centroid	(m10/m00, m01/m00)	Object centre and tracking
Equivalent diameter	$2\sqrt{(A/\pi)}$ converted to mm	Compare scratch or rust-region footprint

2.4 High-level classification

The high-level stage uses a feature vector such as [length, width, aspect ratio, circularity, colour statistics, texture score, OCR confidence and swelling estimate]. An SVM can form a separating boundary between classes; k-NN classifies a sample using the labels of nearby training vectors. A safety rule layer should remain separate from the learned classifier. For example, a pouch cell with swelling above its calibrated limit must be rejected even if the formfactor classifier is confident.

2.5 Comparison of accuracy and latency

The task brief does not supply a labelled test set or measured execution times. The table below therefore defines a measurement plan and expected relative behaviour rather than inventing results. The implementation should test clean cells, touching cells, oil smudges, shadows, reflective glare and partially obscured codes.

Pipeline	Accuracy result to record	Latency result to record	Likely failure case
A – Low-level	Correct / total on labelled test set	Median and 95th percentile milliseconds per image	Shadow or oil stain counted as defect
B – Low + mid	Correct form-factor and defect decisions / total	Contour extraction + feature calculation time	Two touching cells merge into one contour
C – Low + mid + high	Per-class precision, recall, F1 and confusion matrix	Feature extraction + SVM/k-NN + OCR time	Unseen battery design or domain shift

2.6 End-to-end deep learning comparison

A CNN learns spatial filters from labelled images and progressively builds features from edges to textures and object parts. A Vision Transformer (ViT) divides an image into patches and uses attention to model relationships between patches. These approaches can learn complex patterns such as irregular swelling and unfamiliar surface defects, but they normally require a sufficiently large and diverse dataset, model validation and more compute than simple thresholding or contour rules.

On a four-core CPU without a GPU, a compact classical pipeline is easier to time and maintain. A small quantized model may be evaluated later, but the model must be benchmarked on the actual edge PC. Training and test sets must be separated by physical battery or production batch to avoid data leakage.

2.7 Level-annotated pipeline diagram

Level	Block	Output passed forward
Acquisition	Global-shutter camera + synchronized LED strobe	Raw image and exposure metadata
Low level	Denoise → flat-field correction → colour conversion → segmentation	Corrected image and foreground mask
Mid level	Contours → geometry → texture / colour / OCR regions	Feature vector, object location and measurements
High level	SVM / k-NN / OCR confidence + safety rule engine	Form factor, chemistry, defect class and confidence
Control	PLC interlock → pneumatic piston timing	Accept / reject / manual review signal
Audit	Image archive + decision log	Traceable evidence for each cell

Decision: Technique C is preferred as the target architecture. Technique B is a useful interpretable baseline and should be retained as a fallback diagnostic pipeline.

TASK 3 – DIGITAL REPRESENTATION: COLOUR SPACE SELECTION FOR OXIDE RUST ANALYSIS [4 MARKS]

Rust on steel battery jackets often appears brownish-red. The image representation must separate true colour changes from brightness changes caused by ambient illumination, reflections and surface curvature.

3.1 RGB/BGR, HSV and CIE L*a*b* comparison

Technique	Representation	Advantages	Limitations
A – RGB/BGR	Three additive colour channels: red, green and blue	Direct camera output; simple and fast	All channels change with illumination intensity; channel overlap makes rust separation difficult
B – HSV	Hue, Saturation and Value	Hue represents dominant tint; S and V help reject low-saturation glare and very dark regions	Hue becomes unstable when saturation is low; metallic highlights can distort hue
C – CIE L*a*b*	L* lightness, a* green↔red, b* blue↔yellow	Perceptually organized; ΔE compares colour distance from a healthy casing reference	Requires camera colour calibration and controlled illumination; conversion adds computation

3.2 Illumination scaling analysis

The brief specifies three illumination conditions: baseline 1.0×, dim 0.4× and intense 1.8×. In raw RGB, if a pixel channel is approximately proportional to illumination, a baseline channel value of 100 may become approximately 40 under 0.4× light and 180 under 1.8× light, before clipping and camera response effects. Thus a fixed RGB threshold is not stable.

HSV separates brightness into V, while hue can remain comparatively stable for a coloured surface under moderate intensity changes. However, specular reflection reduces saturation and can make hue unreliable. CIE L*a*b* provides a useful ΔE measure after white-balance and colour calibration. For the proposed system, Lab is suitable for colourdistance measurement, while HSV is a computationally efficient secondary representation for segmentation and highlight rejection.

Lighting condition	Relative illumination	RGB response tendency	Expected stable feature
Baseline	1.0×	Reference channel values	Calibrated Lab ΔE and HSV hue/saturation
Dim	0.4×	Approx. 40% of linear-light signal before camera processing	Normalized colour and adequate exposure / strobe energy
Bright	1.8×	Approx. 180%; highlights may clip	Controlled LED lighting, exposure lock and Lab calibration

3.3 ΔE colour distance

For a healthy steel reference colour (L0*, a0*, b0*) and a measured pixel (L*, a*, b*), the simple CIE76 colour difference is:

$$\Delta E^*ab = \sqrt{[(L^* - L0^*)^2 + (a^* - a0^*)^2 + (b^* - b0^*)^2]}.$$

A larger ΔE indicates a larger colour difference from the calibrated healthy casing. A production threshold must be determined from labelled rust and non-rust samples under the actual dome illumination. It should not be selected from a single photograph.

3.4 Image array and memory architecture

A standard colour image loaded by OpenCV is usually represented as a NumPy array with shape (height, width, 3) and dtype uint8. The first index identifies the row, the second identifies the column and the final index identifies the B, G or R channel. A 1920 × 1080 BGR image therefore contains 1920 × 1080 × 3 = 6,220,800 bytes, approximately 5.93 MiB.

3.5 Integer overflow: NumPy addition and cv2.add

An unsigned 8-bit integer can store values from 0 to 255. If ordinary uint8 array addition is used, values may wrap modulo 256. For example, 250 + 20 = 270, but 270 cannot be represented in uint8; the wrapped result is 14. This creates an incorrect dark pixel.

Operation	Input values	Result	Interpretation
uint8 NumPy addition	250 + 20	14 (wrap-around)	Incorrect due to overflow
cv2.add on uint8	250 + 20	255	Saturating addition; clips at upper bound
float32 addition	250.0 + 20.0	270.0	No uint8 overflow; convert / clip deliberately later

For safe arithmetic, convert to a wider type before operations that may exceed the 8-bit range: image.astype(np.float32) or image.astype(np.uint16). Use cv2.add when saturating 8-bit addition is the intended operation.

Decision: Use a calibrated Lab ΔE measure for rust colour comparison, with HSV-based masking to remove lowsaturation highlights and dark background pixels. The final thresholds must be validated across the full 1,500–45,000 lux range.

TASK 4 – TOPOLOGICAL ANALYSIS: PIXEL CONNECTIVITY FOR CASING SCRATCH COUNTING [4 MARKS]

Connectivity determines which foreground pixels belong to the same connected component. The choice changes how diagonal scratches, checkerboard contacts and small noisy regions are counted.

4.1 Four-, eight- and mixed-connectivity

Technique	Neighbourhood	Behaviour	Use / limitation
A – 4-connectivity	North, south, east and west	Diagonal pixels are separate unless joined orthogonally	Avoids some diagonal merging but can fragment a real diagonal scratch
B – 8-connectivity	All eight surrounding pixels	Diagonal contact is treated as connected	Preserves diagonal scratches but may merge objects touching at corners
C – m-connectivity	4-neighbours plus selected diagonal neighbours only when common 4-neighbours do not create ambiguity	Reduces checkerboard loop ambiguity	Useful for defining consistent digital topology; requires custom implementation if not directly available

4.2 Synthetic pattern tests

Three binary test patterns should be used. A diagonal one-pixel line tests whether diagonal continuity is preserved. A checkerboard pattern tests ambiguous diagonal contacts. A noisy background tests whether small isolated components can be removed without deleting genuine narrow scratches.

Pattern	4-connectivity result	8-connectivity result	m-connectivity expectation
Diagonal chain	Several isolated pixels	One connected component	Connected when diagonal connection is unambiguous
Checkerboard joints	Separate orthogonal groups	Many diagonal groups merge	Avoids ambiguous diagonal loops
Noise around scratch	Noise components remain separate	Nearby diagonal noise may merge	Apply area and shape filtering after connectivity

4.3 Distance measures

Metric	Definition for pixel offset (dx, dy)	Geometry
Euclidean (De)	$\sqrt{(dx^2 + dy^2)}$	Straight-line distance; appropriate for physical radial size
City-block (D4)	$\lvert dx\rvert + \lvert dy\rvert$	Manhattan distance; grid-aligned movement only
Chessboard (D8)	$\max(\lvert dx\rvert, \lvert dy\rvert)$	Diagonal and orthogonal steps have equal unit cost

For calibrated physical measurement, the Euclidean metric is the most direct representation of straight-line distance. If the horizontal and vertical pixel scales differ, use calibrated scales sx and sy: $dmm = \sqrt{[(dx \cdot sx)^2 + (dy \cdot sy)^2]}$. This avoids assuming square pixels or equal physical scale in both axes.

4.4 Scratch and rust measurement procedure

1. Convert the corrected image into a suitable colour or intensity representation.
2. Create a candidate defect mask using colour, texture and local contrast.
3. Apply connected-component labelling with a documented connectivity convention.
4. Remove components below a minimum area only after checking that the 0.3 mm defect requirement is still detectable.
5. Calculate component area, perimeter, major-axis length and equivalent diameter.
6. Reject when there are more than two distinct scratches or when a rust component exceeds the 0.5 mm equivalentdiameter rule.

4.5 Custom mixed-connectivity implementation concept

OpenCV connectedComponents supports 4- and 8-connectivity. A mixed-connectivity implementation can be written using a flood-fill or union-find structure. For each foreground pixel, inspect its 4-neighbours first. Inspect a diagonal

neighbour only if the common 4-neighbour pixels do not already create an ambiguous connection. The exact convention must be fixed and tested because different definitions of m-connectivity can produce different labels.

Decision: Use 8-connectivity for preserving narrow diagonal scratches in the candidate mask, followed by geometric filtering and explicit duplicate-merge checks. Use Euclidean calibrated distances for physical dimensions. If mconnectivity is required for the formal analysis, include the custom routine and its chosen definition in the code appendix.

TASK 5 – PHYSICAL ACQUISITION: SENSOR SHUTTER AND EXPOSURE TOPOLOGY [5 MARKS]

A moving battery must be captured with sufficiently short exposure to avoid motion blur. At 1.2 m/s, even a millisecond exposure produces measurable displacement.

5.1 Motion blur calculation

Motion blur distance is calculated by: blur distance = conveyor speed × exposure time.

Exposure	Calculation at 1.2 m/s	Blur distance	Interpretation
4.0 ms	1.2 × 0.004	0.0048 m = 4.8 mm	Too large for micro-crack inspection
1.0 ms	1.2 × 0.001	0.0012 m = 1.2 mm	Still too large for fine defects
0.05 ms	1.2 × 0.00005	0.00006 m = 0.06 mm	Preferred of the three exposure choices

For a 0.3 mm crack, a 0.06 mm blur is 20% of the defect width. This is substantially better than 1.2 mm or 4.8 mm, but the final acceptable blur must be checked against pixel scale, lens MTF and the defect-detection algorithm.

5.2 Rolling shutter, global shutter and line scan

Technique	Acquisition behaviour	Advantages	Risks
A – Rolling shutter + constant light	Rows are exposed/read at slightly different times	Low cost and widely available	Moving objects skew; 100 Hz flicker can produce bands across rows
B – Global shutter + encodertriggered LED strobe	All pixels expose together; short pulse freezes motion	Low geometric distortion and repeatable illumination	Needs synchronization, strobe driver and sufficient peak light
C – Line-scan camera	Captures one line per encoder displacement increment	Excellent for continuous moving web / conveyor imaging	Requires accurate encoder synchronization and uniform line illumination

5.3 Flicker and rolling-shutter distortion

The overhead tube lights operate from a 50 Hz mains supply and may produce 100 Hz brightness modulation. The flicker period is 1/100 = 0.01 s = 10 ms. If row exposure times differ across a significant fraction of this period, different image rows can record different brightness levels. This can create horizontal bands and false colour or texture changes.

A global shutter removes row-to-row exposure timing differences. A synchronized LED strobe also makes the scene illumination repeatable and reduces dependence on daylight. The strobe pulse should be triggered from the belt encoder or a deterministic camera trigger.

5.4 Sensor signal chain

Stage	Physical process	Output
1. Photons	Light reflected from the battery reaches the pixel array	Optical energy
2. Charge integration	Photodiode converts incident photons into accumulated charge	Pixel charge
3. Voltage conversion	Charge is converted to a voltage signal	Analog pixel voltage
4. Analog conditioning	Amplification, correlated double sampling and noise control	Conditioned analog signal
5. ADC digitization	Analog-to-digital converter quantizes the signal	Digital pixel value
6. ISP / host processing	Demosaic, white balance, correction and transfer	Image array for OpenCV

5.5 CCD versus CMOS

Modern industrial CMOS sensors are generally preferred for this application because they support high frame rates, global-shutter variants, low power and direct digital integration. CCD sensors can provide good image quality and uniformity, but high-speed industrial designs are commonly available in CMOS. The final choice should be based on

measured quantum efficiency, read noise, full-well capacity, global-shutter performance and availability of a hardware trigger.

Decision: Technique B is the preferred area-scan solution: global shutter, short synchronized LED strobe and encoder trigger. Technique C is a strong alternative if the conveyor geometry and line-scan optics can be commissioned reliably.

TASK 6 – SPATIAL SAMPLING: RESOLUTION OPTIMIZATION VS DATA THROUGHPUT [5 MARKS]

The horizontal field of view is 320 mm. Spatial scale is therefore calculated by dividing 320 mm by the horizontal pixel count. The smallest required defect is 0.3 mm, and the laser-etched stroke width is 0.4 mm.

6.1 Spatial scale and defect samples

Resolution option	Horizontal pixels	Scale (mm/pixel)	Samples across 0.3 mm crack	Samples across 0.4 mm stroke
VGA 640 × 480	640	320/640 = 0.5000	0.60 px	0.80 px
Full HD 1920 × 1080	1920	320/1920 = 0.1667	1.80 px	2.40 px
12 MP 4096 × 3000	4096	320/4096 = 0.0781	3.84 px	5.12 px

The Nyquist criterion requires at least two samples per repeating feature period as a basic sampling condition. A 0.4 mm text stroke at 0.1667 mm/pixel is represented by about 2.4 pixels and is near the minimum practical range. A 0.3 mm crack at the same scale is represented by only 1.8 pixels, so it is not robustly resolved. The 12 MP option gives about 3.84 pixels across a 0.3 mm defect and provides a more useful margin.

6.2 Data rate and five-day uncompressed storage

Assumption: one 8-bit BGR image per cell, 3 bytes per pixel, 15 cells/s, continuous capture for 5 days. Data rate = width × height × 3 × 15 bytes/s. Five-day storage = data rate × 432,000 seconds.

Resolution	Bytes/image	Data rate at 15 fps	Five-day storage
640 × 480	921,600 B	13.824 MB/s	5.97 TB
1920 × 1080	6,220,800 B	93.312 MB/s	40.31 TB
4096 × 3000	36,864,000 B	552.96 MB/s	238.88 TB

The calculated storage volumes exceed the 1.5 TB SSD capacity by a large margin. Continuous uncompressed retention is therefore impossible. The system must use lossless or visually lossless compression, store selected regions of interest, retain full-resolution images for the required audit period using a larger archive, or reduce the stored frame content while keeping the decision evidence. If the policy literally requires every complete full-resolution image for five days, the storage capacity must be increased.

6.3 Gigabit Ethernet feasibility

Gigabit Ethernet has a theoretical line rate of 1,000 Mbit/s, equivalent to 125 MB/s before protocol overhead. VGA at 13.824 MB/s is comfortable. Full HD at 93.312 MB/s is feasible but leaves limited headroom after packet overhead and other traffic. The 12 MP uncompressed stream at 552.96 MB/s exceeds Gigabit Ethernet capacity and would require compression, a faster interface or reduced frame rate.

6.4 Minimum frame rate to cover a 70 mm cell

A 70 mm long cell moving along the belt occupies 70 mm of the 240 mm vertical FOV. A conservative no-blind-spot condition for frame-to-frame displacement is that the belt should not move more than the remaining vertical FOV after accounting for object length: 240 − 70 = 170 mm. At 1.2 m/s, maximum interval = 0.170/1.2 = 0.1417 s, so the minimum frame rate is 1/0.1417 = 7.06 frames/s. This is only a geometric coverage limit; for 15 cells/s, the camera and trigger system should support at least 15 images/s and preferably a higher rate to handle overlap, vibration and trigger margin.

Decision: VGA is inadequate. Full HD is marginal for 0.3 mm defects. The 12 MP option provides the best spatial detail but cannot stream uncompressed over GigE or fit the five-day archive. A practical design should use an industrial global-shutter sensor with sufficient horizontal pixels (preferably at least 5 MP), ROI acquisition and lossless compression, with archive capacity explicitly sized.

TASK 7 – BRIGHTNESS QUANTIZATION: DYNAMIC RANGE AND DENT DETECTION [4 MARKS]

A dent, pouch deformation or rust pit may create only a 5–8 grey-level difference from the healthy casing on an 8-bit scale. Quantization must preserve these subtle differences while controlling storage and compute cost.

7.1 Bit depth and number of intensity levels

Mode	Bits/channel	Number of levels	Step size relative to 8-bit scale	Assessment
A – Compressed mode	4-bit	16	About 17 levels per 8-bit step	Too coarse for subtle 5–8 level changes
A – Compressed mode	6-bit	64	About 4.05 levels per 8-bit step	May preserve large contrast; small defects become fragile
B – Standard	8-bit	256	1 level	Baseline; preserves the specified 5–8 level difference
C – HDR raw	12-bit	4096	1/16 of an 8-bit step when normalized	Greater highlight/shadow headroom if sensor noise and calibration support it

7.2 Quantization error

For uniform quantization with step size Δ, the ideal quantization error is bounded by approximately $\pm\Delta/2$. Reducing bit depth increases Δ. If the original 8-bit contrast is only 5–8 levels, a 4-bit representation can map healthy and defective values into the same output bin. This can erase the defect contrast or create false contour bands.

7.3 False contouring

A smooth curved battery shoulder contains gradual brightness transitions. When the number of available levels is reduced, many adjacent pixels receive the same quantized value. The resulting flat bands create visible contour boundaries that were not present in the physical object. These artificial boundaries may be mistaken for dents or seams.

7.4 Image bit depth versus model quantization

Concept	What is quantized	Purpose	Potential risk
Image quantization	Pixel intensity values (e.g. 12-bit sensor data to 8-bit image)	Controls image dynamic range and storage	Small intensity differences may be lost
Model quantization	Neural-network weights and activations (e.g. FP32 to INT8)	Reduces memory and accelerates inference	Accuracy may decrease if calibration is poor

7.5 Storage trade-off

For the same pixel count, 12-bit packed raw data requires 1.5 times the storage of 8-bit data; if stored in a 16-bit container, it may require twice the storage. Under the 1.5 TB archive limit, preserving every pixel at 12-bit depth is expensive. However, reducing spatial resolution can destroy a 0.3 mm crack completely. The correct optimization order is to preserve enough spatial sampling first, then select the lowest bit depth that retains the measured defect contrast with margin.

Decision: Use 8-bit for the initial balanced system only if controlled lighting and sensor testing prove adequate contrast. Prefer 10–12-bit acquisition where the sensor supports it, then retain calibrated high-bit-depth defect ROIs and use a compressed 8-bit review image for routine audit.

TASK 8 – IMAGE FORMATION: GEOMETRIC PROJECTION AND PHOTOMETRIC CORRECTIONS [5 MARKS]

Image formation depends on the lens, sensor, camera distance, object height and illumination. A flat conveyor calibration does not automatically correct the apparent size of objects elevated above the belt.

8.1 Focal-length calculation

For a rectilinear lens, horizontal field of view is approximated by FOV = sensor width × distance / focal length. Rearranging gives focal length = sensor width × distance / FOV.

Using a representative 1/2-inch sensor active width of 6.4 mm, camera distance 400 mm and required FOV 320 mm:

$f = (6.4 \times 400) / 320 = 8.0$ mm.

This is an approximate first-order result. The actual active sensor width, lens distortion, focus distance and principal point must be taken from the camera/lens datasheets and calibration images.

8.2 Perspective projection error

The pinhole projection equation is $x = fX/Z$. A calibration made at the belt plane uses $Z = 400$ mm. A point on the top of a 30 mm-high battery is at $Z = 400 - 30 = 370$ mm. Its image magnification relative to the belt-plane calibration is $400/370 = 1.0811$, or approximately 8.11% larger.

For a 70 mm dimension located at the top surface, the apparent dimension under the belt-plane scale is approximately $70 \times 1.0811 = 75.68$ mm. The apparent error is about 5.68 mm. This is much larger than the required ±0.5 mm swelling precision.

8.3 Lens and lighting configurations

Technique	Configuration	Benefits	Limitations
A – Standard perspective + ambient light	Fixed focal lens; open factory lighting	Lowest cost and simple installation	Perspective error, variable illumination and glare
B – Standard lens + diffuse LED dome + calibration	Enclosed diffuse lighting, checkerboard calibration and flatfield correction	Stable illumination and corrected lens geometry	Perspective remains heightdependent unless height is measured or constrained
C – Telecentric lens + diffuse dome	Object-space telecentric optics and controlled light	Near-constant magnification over working distance	High lens cost, limited FOV and mechanical constraints

8.4 Flat-field correction

A flat-field image estimates the illumination response across the field. Let $I(x,y)$ be the captured image, $D(x,y)$ the dark reference and $F(x,y)$ the flat-field reference. A corrected image can be calculated as: $I_{corrected}(x,y) = [I(x,y) - D(x,y)] \times K / [F(x,y) - D(x,y) + \varepsilon]$, where K is a normalization constant and ε prevents division by zero. This reduces vignetting and spatial illumination gradients. The flat-field reference must be captured with the same lens aperture, lighting and camera settings used in production.

8.5 Homomorphic correction concept

An image can be modelled as $f(x,y) = i(x,y) \times r(x,y)$, where i is illumination and r is reflectance. Taking logarithms gives $\log f = \log i + \log r$. Filtering the low-frequency illumination component and recombining it with the reflectance component can reduce gradual shading. This method should not be used as a substitute for stable lighting when the object has strong specular reflections.

Decision: Use a calibrated standard lens with a diffuse LED dome for the budget-conscious design, but measure battery height and apply height compensation. If ±0.5 mm dimensional precision must be guaranteed across varying heights, telecentric optics or a separate height/depth measurement channel may be necessary.

TASK 9 – SYSTEMS INTEGRATION: BALANCED FRONT-END OPTIMIZATION CHALLENGE [6 MARKS]

This task combines the sensor, optics, illumination, processing, archive and actuator into a complete system. The available hardware budget is 1,25,000, and the decision must reach the rejecter within 400 ms.₹

9.1 Three system packages

Package	Main components	Advantages	Risks / constraints
A – Budget	1.3 MP rolling-shutter USB camera, ambient light, JPEG, low+mid processing	Lowest purchase cost and easy setup	Motion skew, flicker, lossy evidence and weaker micro-defect sampling
B – Balanced	2 MP global-shutter GigE camera, LED dome/strobe, calibrated standard lens, PNG, three-level pipeline	Stable geometry, controlled light and interpretable processing	2 MP is marginal for 0.3 mm defects across 320 mm FOV; archive still needs planning
C – Premium	4K line-scan, telecentric lens, line light, 12-bit raw and three-level pipeline	Best spatial detail and continuous motion capture	High cost, complex synchronization and very high raw data rate

9.2 Illustrative budget allocation

The following is a planning allocation, not a vendor quotation. Final procurement must use current supplier quotes and verify interface, sensor size, lens mount, strobe compatibility and warranty.

Component	Planning allowance ( )₹	Reason
Global-shutter industrial camera (2–5 MP class)	38,000	Primary imaging sensor; prefer ≥5 MP if available within budget
Calibrated C-mount lens	9,000	FOV and working-distance match
Diffuse LED dome / ring + strobe controller	18,000	Stable illumination and short exposure
Encoder / trigger interface and cables	7,000	Repeatable capture synchronization
Industrial PC interface / I/O / PLC relay integration	12,000	Connects vision decision to actuator
Protective enclosure and mounting	8,000	Blocks ambient variation and vibration
SSD / archive provision	10,000	Local buffering and audit records
Mechanical and commissioning allowance	13,000	Mounting, calibration and integration
Contingency	10,000	Unplanned adapters, wiring and replacement items
Total planning allowance	1,25,000	Within the stated cap

9.3 End-to-end data and control path

Stage	Operation	Target time budget
1	Encoder detects cell position and triggers camera	5 ms
2	Global-shutter exposure with LED strobe	0.05 ms exposure; trigger overhead included in acquisition
3	Image transfer and buffering	20 ms
4	Flat-field correction, denoising and segmentation	35 ms
5	Geometry, colour, defect and OCR feature extraction	70 ms
6	Classifier and safety rule evaluation	35 ms
7	PLC message, interlock and decision commit	15 ms
8	Reserve for scheduling / network jitter	50 ms
Total planned decision budget	End-to-end processing target	230 ms

The planned 230 ms budget is below the 400 ms physical deadline and leaves 170 ms margin. The actuator timing must still be tested at the real 0.48 m camera-to-piston distance. At 1.2 m/s, the travel time is exactly 0.48/1.2 = 0.4 seconds.

9.4 Engineering decision matrix

Condition	Budget A	Balanced B	Premium C

Changing daylight / storm clouds	Poor; ambient light changes image	Strong; enclosed controlled LED	Strong; controlled line light
Fast movement	Rolling-shutter skew risk	Global shutter freezes frame	Line-scan synchronized to encoder
Specular glare	High risk	Dome reduces directional reflection	Can be controlled with specialized line lighting / polarization
0.3 mm defect	Likely under-sampled	2 MP is marginal	Best sampling
400 ms deadline	Likely possible but variable	Suitable with bounded pipeline	Possible, but integration complexity
Budget compliance	Likely	Most realistic	High risk of exceeding budget
Maintenance complexity	Low to medium	Medium	High

9.5 Storage and traceability design

The archive should store a compressed full-frame or selected high-resolution crop, a thumbnail, OCR text, defect mask, measurements, confidence values, timestamp and final actuator command. A ring buffer can retain pre-trigger and post-trigger frames around each cell. If policy requires the complete image for every cell for five days, the 1.5 TB SSD is insufficient for uncompressed capture at the calculated rates; management must approve additional storage or a compliant compression/retention architecture.

Decision: Package B is the best cost-balanced direction, but the 2 MP specification should be upgraded to at least a 5 MP global-shutter sensor if the 0.3 mm defect must be resolved across the full 320 mm field. Do not claim the ≤1.5% miss-rate requirement until validated using representative, independently labelled battery samples.

TASK 10 – ADAPTIVE ENGINEERING: HIGH-SPEED PLANT SCALE-UP CHALLENGE [8 MARKS]

Three months after deployment, the line speed doubles from 1.2 m/s to 2.4 m/s and throughput increases from 15 to 30 cells/s. The decision deadline is reduced to 200 ms. A new requirement adds detection of internal short-circuit risks through subtle 0.5°C thermal anomalies in the MWIR spectrum.

10.1 Recalculate motion blur at 2.4 m/s

The maximum exposure for a selected allowable blur b is t = b/v. At 2.4 m/s, the same exposure produces twice the blur distance as at 1.2 m/s.

Exposure	Calculation	Blur distance at 2.4 m/s	Implication
4.0 ms	2.4 × 0.004	9.6 mm	Unacceptable for fine defect inspection
1.0 ms	2.4 × 0.001	2.4 mm	Unacceptable
0.05 ms	2.4 × 0.00005	0.12 mm	Much better; still a significant fraction of 0.3 mm defect

If the design limits motion blur to 0.06 mm, the maximum exposure is 0.00006/2.4 = 0.000025 s = 25 µs. If 0.10 mm blur is acceptable, the maximum exposure is 41.7 µs. Therefore, the new system should use an encoder-synchronized strobe with an exposure target around 25–40 µs, subject to sensor sensitivity and strobe power.

10.2 Recalculate spatial and data requirements

The spatial scale of a camera does not change merely because the belt speed doubles. If the same FOV and pixel count are retained, mm/pixel remains unchanged. However, the frame rate, strobe repetition rate and data throughput must double to capture 30 cells/s.

Resolution	mm/pixel across 320 mm	Samples across 0.3 mm	BGR data rate at 30 fps	Five-day uncompressed storage
640 × 480	0.5000	0.60	27.648 MB/s	11.94 TB
1920 × 1080	0.1667	1.80	186.624 MB/s	80.62 TB
4096 × 3000	0.0781	3.84	1,105.92 MB/s	477.76 TB

10.3 Adaptation strategies

Strategy	Changes	Benefits	Limitations
A – Retain + ROI optimization	Keep current hardware; shorten exposure; crop ROIs; downsample non-critical regions	Lowest cost and quickest change	Cannot create missing spatial detail or detect invisible thermal anomalies with visible sensor
B – Modify sensing and shutter	Higher frame-rate global shutter,	Improves visible imaging speed	Still cannot detect MWIR-only

	ultra-short strobe, light tunnel and INT8 arrays	and ambient rejection	thermal signatures
C – Redesign architecture	Visible line-scan plus synchronized MWIR thermal sensor in regulated enclosure; dual-stream fusion	Addresses visible defects and thermal anomalies	Highest integration and calibration complexity; cost may exceed original budget

10.4 MWIR thermal sensing

A visible camera measures reflected visible light. It cannot reliably observe a 0.5°C internal thermal anomaly that is only detectable in the MWIR spectrum. A dedicated MWIR thermal sensor is therefore required. The sensor must be selected for its spectral band, thermal sensitivity (NETD), spatial resolution, frame rate, lens transmission and calibration stability. An uncooled microbolometer is more commonly associated with long-wave infrared; MWIR implementations are specialized and may require cooled detectors. The exact sensor technology must be verified against the required MWIR band and 0.5°C sensitivity.

The thermal channel should be housed in a regulated enclosure to reduce drift. A reference source or periodic nonuniformity correction may be needed. Visible and thermal frames must be synchronized and geometrically registered so that a thermal hotspot is assigned to the correct battery.

10.5 Few-shot learning and foundation models

A pretrained vision foundation model may help classify new thermal defect shapes using a small number of labelled examples. Few-shot learning uses a small support set to compare new samples with learned visual representations. This can reduce the number of labelled images required for initial adaptation, but it does not eliminate validation. Thermal images differ from ordinary RGB images, so a model pretrained on visible photographs may suffer severe domain shift.

Risk	Why it matters	Control measure
Domain shift	Thermal texture and contrast differ from visible images and training data	Collect MWIR data from actual cells across operating temperatures and battery chemistries
Inference latency	Large foundation models may exceed 200 ms on a 4-core CPU	Use compact embeddings, ROI inference, quantization and measured CPU benchmarks
Traceability	A model update can change decisions without obvious rule changes	Version models, store input crops, confidence, calibration and software hashes
Few-shot overfitting	A few examples may not represent production variation	Use independent validation batches and uncertainty-based reject logic
Thermal calibration drift	Sensor temperature and enclosure changes alter measurements	Reference checks, non-uniformity correction and periodic calibration

10.6 200 ms timing budget

Stage	Target time
Encoder trigger + exposure	2 ms (exposure itself targeted at 25–40 µs)
Visible and thermal frame transfer / synchronization	25 ms
Visible correction and segmentation	25 ms
Thermal correction and hotspot extraction	25 ms
Feature fusion and classification	45 ms
Safety rule engine and confidence gate	15 ms
PLC commit and actuator scheduling	10 ms
Jitter reserve	30 ms
Total	177 ms

The 177 ms allocation is a design target, not a measured result. It leaves 23 ms margin against the 200 ms deadline. If the measured 95th-percentile latency exceeds 200 ms, the system must reduce computation, move processing closer to the camera, or change the control architecture.

10.7 RETAIN, MODIFY or REDESIGN decision

Strategy A cannot satisfy the new requirement because no visible-light image-processing optimization can reveal a thermal-only anomaly. Strategy B improves visible capture and timing but still lacks the required MWIR sensor. Strategy C is therefore the technically correct architecture decision: redesign as a synchronized dual-stream visible + MWIR system. The four-core CPU constraint makes a compact thermal feature pipeline and carefully benchmarked

model necessary. A large foundation model should not be placed directly in the critical path unless it demonstrates the required worst-case latency.

Decision: REDESIGN. Retain the existing visible camera only as a secondary channel if it meets the new spatial and exposure requirements; add a dedicated MWIR sensing path, synchronization, thermal calibration and a bounded fusion pipeline. Re-open the budget because the new thermal requirement is a fundamental sensor change, not a software-only upgrade.

OVERALL ENGINEERING DECISION AND CONCLUSION

The battery-sorting line requires a non-contact machine vision system because manual inspection cannot reliably sustain 15 cells/s while reading small chemistry labels and detecting subtle safety defects. Image-processing assistance improves visibility but does not remove the human decision bottleneck. A three-level automated pipeline provides the best foundation: low-level correction and segmentation, mid-level geometry and defect features, and high-level classification/OCR with explicit safety rules.

The principal design challenge is not only classification accuracy. It is the combination of spatial sampling, motion blur, changing illumination, reflective surfaces, storage limits and actuator timing. A 0.3 mm defect across a 320 mm field requires substantially more than VGA resolution. Full HD is marginal, while 12 MP offers better sampling but creates excessive uncompressed data. A global-shutter camera with synchronized strobe lighting, calibrated optics and region-of-interest processing is the most practical visible-light starting point.

The focal-length estimate for a 1/2-inch sensor with 6.4 mm active width is approximately 8 mm. A 30 mm object height creates approximately 8.11% scale difference relative to a belt-plane calibration, showing why height compensation or telecentric optics is important for ±0.5 mm dimensional measurement. At 1.2 m/s, a 0.05 ms exposure creates 0.06 mm blur; at 2.4 m/s it creates 0.12 mm blur. The scale-up therefore requires an exposure near 25–40 µs for fine inspection.

For the original line, the balanced global-shutter and controlled-light package is the best direction, but its sensor resolution must be selected to meet the 0.3 mm defect requirement and the archive policy must be reconciled with the 1.5 TB capacity. For the scaled line, a visible-only solution is insufficient because the new thermal anomaly is invisible in visible light. The correct decision is REDESIGN with synchronized visible and MWIR sensing, thermal calibration and a measured sub-200 ms processing path.

FINAL RECOMMENDATION SUMMARY

Design element	Recommended choice	Reason
Operational workflow	Automated computer vision with manual exception review	15 cells/s and safety-critical sorting
Visible sensor	Global-shutter industrial CMOS	Low distortion and deterministic timing
Illumination	Enclosed diffuse LED dome / strobe synchronized to encoder	Controls daylight and 100 Hz flicker
Colour analysis	Calibrated Lab ΔE plus HSV masking	Robust rust colour comparison
Connectivity / shape	8-connectivity plus Euclidean calibrated geometry	Preserves diagonal scratches and physical dimensions
Pipeline	Low + mid + high-level classical CV / ML with safety rules	Interpretable and CPU-efficient
Archive	Compressed evidence + ROI + metadata; verify 5-day policy capacity	Uncompressed archive exceeds 1.5 TB
Scale-up	Dual-stream visible + MWIR redesign	Required for 0.5°C thermal anomaly detection

APPENDIX A – CORE IMPLEMENTATION EXAMPLES

The following code fragments show the core operations required by the proposed design. They are implementation templates and must be integrated, tested and benchmarked on the actual camera and edge PC. They do not represent completed experimental measurements.

A.1 Safe image arithmetic and colour conversion

```
import cv2 import numpy
as np

# Read a colour image (OpenCV uses BGR channel order). image =
cv2.imread("battery_sample.png")

# Convert to a wider type before arithmetic that may exceed 255. wide = image.astype(np.uint16)
added = np.clip(wide + 20, 0, 255).astype(np.uint8)
```

```
# Convert to HSV and Lab for complementary colour analysis. hsv =
cv2.cvtColor(image, cv2.COLOR_BGR2HSV) lab = cv2.cvtColor(image,
cv2.COLOR_BGR2LAB)

# Example: remove very low-saturation highlight/background pixels. mask = ((hsv[:, :, 1] > 35) & (hsv[:, :,
2] > 25)).astype(np.uint8) * 255
```

A.2 Flat-field correction

```
import cv2 import numpy
as np

def flat_field_correct(image, dark, flat):
    image_f = image.astype(np.float32)     dark_f =
dark.astype(np.float32)     flat_f = flat.astype(np.float32)

    denominator = flat_f - dark_f     denominator =
np.maximum(denominator, 1.0)

    # Normalize using the mean valid flat-field response.     valid = denominator > 1.0
scale = np.mean(denominator[valid])
    corrected = (image_f - dark_f) * scale / denominator     return np.clip(corrected, 0,

255).astype(np.uint8)
```

A.3 Connected components and calibrated equivalent diameter

```
import cv2 import numpy
as np

def measure_components(binary_mask, mm_per_pixel):     count, labels, stats, centroids =
cv2.connectedComponentsWithStats(         binary_mask, connectivity=8     )

    results = []     for label in range(1, count):         area_px = stats[label,
cv2.CC_STAT_AREA]         area_mm2 = area_px * (mm_per_pixel **
2)
        equivalent_diameter_mm = 2.0 * np.sqrt(area_mm2 / np.pi)

        results.append({
            "area_pixels": int(area_px),
            "area_mm2": float(area_mm2),
            "equivalent_diameter_mm": float(equivalent_diameter_mm),
            "centroid": centroids[label].tolist()
        })     return results
```

A.4 Motion blur calculator

```
def motion_blur_mm(speed_m_per_s, exposure_seconds):     return speed_m_per_s *
exposure_seconds * 1000.0

for exposure in [0.004, 0.001, 0.00005]:     print(exposure, motion_blur_mm(1.2,
exposure), "mm")
```

A.5 End-to-end decision pseudocode

```
def inspect_cell(frame, calibration, model, ocr, limits):     corrected =
apply_flat_field(frame, calibration)     mask = segment_battery(corrected)

    geometry = extract_geometry(mask, calibration)     colour_features =
extract_lab_hsv_features(corrected, mask)     defects = detect_rust_dents_scratches(corrected, mask,
calibration)     chemistry, ocr_confidence = ocr.read_code(corrected, mask)
```

```
    form_factor, class_confidence = model.predict(
        geometry + colour_features     )

    unsafe = defects.has_critical_defect
    unknown_chemistry = ocr_confidence < limits.min_ocr_confidence     uncertain_class =
class_confidence < limits.min_class_confidence

    reject = unsafe or unknown_chemistry or uncertain_class
    return {
        "form_factor": form_factor,
        "chemistry": chemistry,
        "defects": defects,
        "reject": reject
    }
```

APPENDIX B – VERIFICATION CHECKLIST AND VIVA POINTS

B.1 Commissioning checklist

- ☐ Verify camera active sensor width and lens focal length from datasheets.
- ☐ Calibrate lens distortion using a checkerboard at the conveyor plane.
- ☐ Measure the height range of cylindrical and pouch batteries.
- ☐ Verify scale compensation or telecentric measurement for raised surfaces.
- ☐ Confirm global-shutter exposure and strobe synchronization with the encoder.
- ☐ Measure actual exposure duration and motion blur using a moving test target.
- ☐ Test under dim, nominal and bright ambient conditions.
- ☐ Test reflective steel and foil surfaces with and without polarization.
- ☐ Create labelled samples for LCO, LFP, NMC and NiMH chemistry codes.
- ☐ Measure OCR character error rate and confidence calibration.
- ☐ Validate 0.3 mm defects with a traceable reference target.
- ☐ Validate pouch swelling measurement against a dimensional reference.
- ☐ Test touching batteries, partial occlusion, oil smudges, shadows and glare.
- ☐ Calculate inter-rater and intra-rater Cohen’s kappa from real operator trials.
- ☐ Measure median, 95th percentile and worst-case end-to-end latency.
- ☐ Verify PLC interlocks and fail-safe reject behaviour.
- ☐ Check archive capacity against five days of production at maximum throughput.
- ☐ Store image, ROI, metadata, model version and final actuator decision.
- ☐ Run an independent validation set and report false accept / false reject rates.
- ☐ Repeat calibration after maintenance or camera/lens movement.

B.2 Viva questions with short answers

Why is a global shutter preferred? All pixels expose at the same time, reducing geometric skew for moving batteries.

What is motion blur? The distance an object moves during exposure; blur = speed × exposure time.

Why is 0.05 ms exposure useful at 1.2 m/s? It produces 0.06 mm of motion blur, much lower than 1 ms or 4 ms.

Why is HSV useful? It separates hue from saturation and brightness, helping colour-based segmentation.

Why use Lab ΔE? It provides a colour-distance measure from a calibrated healthy reference.

What is connected-component labelling? It groups foreground pixels into separate connected regions.

What is the difference between 4- and 8-connectivity? 4-connectivity uses orthogonal neighbours; 8-connectivity also includes diagonals.

What is the Nyquist sampling principle? At least two samples are required across a repeating feature as a basic sampling limit.

Why is 2 MP marginal for 0.3 mm defects? At 320 mm FOV it gives about 0.1667 mm/pixel, or only 1.8 pixels across 0.3 mm.

What is flat-field correction? A correction that reduces spatial illumination variation using a reference flat image.

What is Cohen's kappa? A statistic that measures categorical agreement after accounting for chance agreement.

Why is a thermal camera required after scale-up? The new 0.5°C internal anomaly is specified as detectable only in the MWIR spectrum.

Why can a foundation model fail on thermal images? Its training domain may differ from thermal imagery, causing domain shift.

Why should low-confidence OCR be rejected? A wrong chemistry label can create a hazardous mixed battery batch.

What does traceability mean in this system? Each decision can be linked to its image, measurements, timestamp, model version and actuator command.

B.3 Evidence that must be attached after actual execution

Evidence	Required content	Status
Camera calibration	Checkerboard images, intrinsic matrix, distortion coefficients and reprojection error	To be measured
Motion test	Encoder speed, exposure duration and measured blur	To be measured
Colour validation	Healthy/rust Lab references and ΔE distributions across illumination	To be measured
Connectivity test	Synthetic pattern masks and component counts for N4, N8 and m-connectivity	To be generated
Classifier validation	Confusion matrix, precision, recall, F1 and perclass support	To be measured
OCR validation	Character error rate for all chemistry codes	To be measured
Latency benchmark	Median, p95 and maximum end-to-end latency on 4-core PC	To be measured
Archive test	Actual bytes per image and five-day storage estimate	To be measured
Safety validation	False accept rate for hazardous / defective batteries	To be measured

REFERENCES / TECHNICAL BASIS

1. Assignment brief supplied for ITA05 – Computer Vision, Assignment 1 – CO1: Vision-Based E-Waste Battery Sorting Line.
2. OpenCV documentation: image filtering, colour-space conversion, thresholding, contours, connected components and geometric measurement functions.
3. NumPy documentation: ndarray data types, array arithmetic and integer overflow behaviour.
4. Standard pinhole camera projection model: x = fX/Z.
5. Sampling theory: Nyquist criterion for representing spatial features.
6. Cohen, J. (1960). A coefficient of agreement for nominal scales. Educational and Psychological Measurement.

Note: Component prices in the planning table are budget allowances only, not verified supplier quotations. All hardware selection and performance claims must be confirmed through datasheets, quotations and on-line validation tests.

SUPPLEMENTARY TECHNICAL CHAPTER 1: DETAILED REQUIREMENTS ENGINEERING

This chapter expands the engineering treatment of detailed requirements engineering for the vision-based e-waste battery sorting line. The discussion is organised around system requirements, stakeholder needs, functional requirements, non-functional requirements, acceptance criteria. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which detailed requirements engineering will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In detailed requirements engineering, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to detailed requirements engineering. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so detailed requirements engineering should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For detailed requirements engineering, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Engineering checklist

- Define measurable requirements for detailed requirements engineering and record all assumptions.
- Check that the selected method is compatible with the conveyor speed and available processing time.
- Test normal, damaged, reflective, partially occluded and low-quality examples.

- Record per-class results and investigate every safety-relevant error.
- Provide a reject or manual-review route for uncertain outcomes.
- Store image IDs, timestamps, configuration and software/model versions.
- Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for detailed requirements engineering. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 2: BATTERY FORM-FACTOR TAXONOMY

This chapter expands the engineering treatment of battery form-factor taxonomy for the vision-based e-waste battery sorting line. The discussion is organised around 18650 cylindrical cells, 21700 cylindrical cells, prismatic cells, pouch cells, dimensions, geometry and visual cues. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which battery form-factor taxonomy will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In battery form-factor taxonomy, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to battery form-factor taxonomy. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so battery form-factor taxonomy should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For battery form-factor taxonomy, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Engineering checklist

- Define measurable requirements for battery form-factor taxonomy and record all assumptions.
- Check that the selected method is compatible with the conveyor speed and available processing time.
- Test normal, damaged, reflective, partially occluded and low-quality examples.
- Record per-class results and investigate every safety-relevant error.
- Provide a reject or manual-review route for uncertain outcomes.
- Store image IDs, timestamps, configuration and software/model versions.
- Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for battery form-factor taxonomy. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 3: BATTERY CHEMISTRY TAXONOMY

This chapter expands the engineering treatment of battery chemistry taxonomy for the vision-based e-waste battery sorting line. The discussion is organised around LCO, LFP, NMC, NiMH, label conventions, ambiguity, chemistry verification and safety implications. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which battery chemistry taxonomy will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In battery chemistry taxonomy, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to battery chemistry taxonomy. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so battery chemistry taxonomy should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For battery chemistry taxonomy, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Engineering checklist

- Define measurable requirements for battery chemistry taxonomy and record all assumptions.
- Check that the selected method is compatible with the conveyor speed and available processing time.
- Test normal, damaged, reflective, partially occluded and low-quality examples.
- Record per-class results and investigate every safety-relevant error.
- Provide a reject or manual-review route for uncertain outcomes.
- Store image IDs, timestamps, configuration and software/model versions.
- Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for battery chemistry taxonomy. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 4: IMAGE ACQUISITION ARCHITECTURE

This chapter expands the engineering treatment of image acquisition architecture for the vision-based e-waste battery sorting line. The discussion is organised around camera placement, lens selection, field of view, working distance, exposure, trigger timing and image transfer. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which image acquisition architecture will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In image acquisition architecture, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to image acquisition architecture. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so image acquisition architecture should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For image acquisition architecture, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was

needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

- Define measurable requirements for image acquisition architecture and record all assumptions.
- Check that the selected method is compatible with the conveyor speed and available processing time.
- Test normal, damaged, reflective, partially occluded and low-quality examples.
- Record per-class results and investigate every safety-relevant error.
- Provide a reject or manual-review route for uncertain outcomes.
- Store image IDs, timestamps, configuration and software/model versions.
- Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for image acquisition architecture. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 5: CAMERA SENSOR FUNDAMENTALS

This chapter expands the engineering treatment of camera sensor fundamentals for the vision-based e-waste battery sorting line. The discussion is organised around pixel array, pixel pitch, quantum efficiency, dynamic range, read noise, bit depth and sensor selection. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which camera sensor fundamentals will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In camera sensor fundamentals, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to camera sensor fundamentals. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so camera sensor fundamentals should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For camera sensor fundamentals, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Engineering checklist

- Define measurable requirements for camera sensor fundamentals and record all assumptions.
- Check that the selected method is compatible with the conveyor speed and available processing time.
- Test normal, damaged, reflective, partially occluded and low-quality examples.
- Record per-class results and investigate every safety-relevant error.
- Provide a reject or manual-review route for uncertain outcomes.
- Store image IDs, timestamps, configuration and software/model versions.
- Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for camera sensor fundamentals. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 6: LENS AND OPTICAL DESIGN

This chapter expands the engineering treatment of lens and optical design for the vision-based e-waste battery sorting line. The discussion is organised around focal length, aperture, depth of field, distortion, focus, magnification and lens mounting. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which lens and optical design will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In lens and optical design, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidencescore meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to lens and optical design. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so lens and optical design should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For lens and optical design, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to

Define measurable requirements for lens and optical design and record all assumptions.
Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for lens and optical design. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 7: ILLUMINATION ENGINEERING

This chapter expands the engineering treatment of illumination engineering for the vision-based e-waste battery sorting line. The discussion is organised around diffuse lighting, coaxial lighting, dark-field lighting, polarisation, shadows, glare and flicker control. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which illumination engineering will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In illumination engineering, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to illumination engineering. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so illumination engineering should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For illumination engineering, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the

design. Calibration records and representative images are particularly valuable because they allow another engineer to Define measurable requirements for illumination engineering and record all assumptions.

Check that the selected method is compatible with the conveyor speed and available processing time.

Test normal, damaged, reflective, partially occluded and low-quality examples.

Record per-class results and investigate every safety-relevant error.

Provide a reject or manual-review route for uncertain outcomes.

Store image IDs, timestamps, configuration and software/model versions.

Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for illumination engineering. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 8: IMAGE FORMATION MODEL

This chapter expands the engineering treatment of image formation model for the vision-based e-waste battery sorting line. The discussion is organised around scene radiance, illumination, reflectance, optics, sampling, quantisation and digital image representation. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which image formation model will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In image formation model, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidencescore meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to image formation model. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so image formation model should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For image formation model, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the

design. Calibration records and representative images are particularly valuable because they allow another engineer to Define measurable requirements for image formation model and record all assumptions.

Check that the selected method is compatible with the conveyor speed and available processing time.

Test normal, damaged, reflective, partially occluded and low-quality examples.

Record per-class results and investigate every safety-relevant error.

Provide a reject or manual-review route for uncertain outcomes.

Store image IDs, timestamps, configuration and software/model versions.

Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for image formation model. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 9: SPATIAL RESOLUTION ANALYSIS

This chapter expands the engineering treatment of spatial resolution analysis for the vision-based e-waste battery sorting line. The discussion is organised around object-space resolution, pixels per millimetre, minimum defect size and sampling limitations. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which spatial resolution analysis will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In spatial resolution analysis, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to spatial resolution analysis. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so spatial resolution analysis should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For spatial resolution analysis, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the

design. Calibration records and representative images are particularly valuable because they allow another engineer to Define measurable requirements for spatial resolution analysis and record all assumptions.

Check that the selected method is compatible with the conveyor speed and available processing time.

Test normal, damaged, reflective, partially occluded and low-quality examples.

Record per-class results and investigate every safety-relevant error.

Provide a reject or manual-review route for uncertain outcomes.

Store image IDs, timestamps, configuration and software/model versions.

Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for spatial resolution analysis. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 10: TEMPORAL RESOLUTION AND MOTION

This chapter expands the engineering treatment of temporal resolution and motion for the vision-based e-waste battery sorting line. The discussion is organised around frame rate, exposure time, conveyor velocity, motion blur, trigger jitter and rolling-shutter effects. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which temporal resolution and motion will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In temporal resolution and motion, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to temporal resolution and motion. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so temporal resolution and motion should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For temporal resolution and motion, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Engineering checklist

- Define measurable requirements for temporal resolution and motion and record all assumptions.
- Check that the selected method is compatible with the conveyor speed and available processing time.
- Test normal, damaged, reflective, partially occluded and low-quality examples.
- Record per-class results and investigate every safety-relevant error.
- Provide a reject or manual-review route for uncertain outcomes.
- Store image IDs, timestamps, configuration and software/model versions.
- Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for temporal resolution and motion. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 11: IMAGE CALIBRATION PROCEDURE

This chapter expands the engineering treatment of image calibration procedure for the vision-based e-waste battery sorting line. The discussion is organised around checkerboard calibration, intrinsic parameters, extrinsic parameters, lens distortion and calibration verification. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which image calibration procedure will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In image calibration procedure, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to image calibration procedure. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so image calibration procedure should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For image calibration procedure, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was

needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Define measurable requirements for image calibration procedure and record all assumptions.
Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for image calibration procedure. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 12: COLOUR REPRESENTATION

This chapter expands the engineering treatment of colour representation for the vision-based e-waste battery sorting line. The discussion is organised around RGB, grayscale, HSV, Lab, colour constancy, white balance and chemistrylabel colour cues. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which colour representation will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In colour representation, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidencescore meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to colour representation. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so colour representation should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manualreview container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For colour representation, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Engineering checklist

- Define measurable requirements for colour representation and record all assumptions.
- Check that the selected method is compatible with the conveyor speed and available processing time.
- Test normal, damaged, reflective, partially occluded and low-quality examples.
- Record per-class results and investigate every safety-relevant error.
- Provide a reject or manual-review route for uncertain outcomes.
- Store image IDs, timestamps, configuration and software/model versions.
- Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for colour representation. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 13: PRE-PROCESSING PIPELINE

This chapter expands the engineering treatment of pre-processing pipeline for the vision-based e-waste battery sorting line. The discussion is organised around image validation, resizing, denoising, contrast normalisation, colour correction and region-of-interest extraction. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which pre-processing pipeline will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In pre-processing pipeline, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidencescore meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to preprocessing pipeline. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so pre-processing pipeline should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For pre-processing pipeline, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was

needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

- Define measurable requirements for pre-processing pipeline and record all assumptions.
- Check that the selected method is compatible with the conveyor speed and available processing time.
- Test normal, damaged, reflective, partially occluded and low-quality examples.
- Record per-class results and investigate every safety-relevant error.
- Provide a reject or manual-review route for uncertain outcomes.
- Store image IDs, timestamps, configuration and software/model versions.
- Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for pre-processing pipeline. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 14: NOISE MODELS AND REDUCTION

This chapter expands the engineering treatment of noise models and reduction for the vision-based e-waste battery sorting line. The discussion is organised around Gaussian noise, Poisson noise, impulse noise, sensor noise, median filtering and Gaussian filtering. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which noise models and reduction will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In noise models and reduction, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to noise models and reduction. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so noise models and reduction should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For noise models and reduction, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Engineering checklist

- Define measurable requirements for noise models and reduction and record all assumptions.
- Check that the selected method is compatible with the conveyor speed and available processing time.
- Test normal, damaged, reflective, partially occluded and low-quality examples.
- Record per-class results and investigate every safety-relevant error.
- Provide a reject or manual-review route for uncertain outcomes.
- Store image IDs, timestamps, configuration and software/model versions.
- Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for noise models and reduction. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 15: CONTRAST ENHANCEMENT

This chapter expands the engineering treatment of contrast enhancement for the vision-based e-waste battery sorting line. The discussion is organised around histogram analysis, histogram equalisation, CLAHE, gamma correction and clipping risks. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which contrast enhancement will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In contrast enhancement, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidencescore meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to contrast enhancement. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so contrast enhancement should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manualreview container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For contrast enhancement, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to

Define measurable requirements for contrast enhancement and record all assumptions.
Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for contrast enhancement. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 16: THRESHOLDING METHODS

This chapter expands the engineering treatment of thresholding methods for the vision-based e-waste battery sorting line. The discussion is organised around global thresholding, adaptive thresholding, Otsu thresholding, illumination variation and binary-mask cleanup. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which thresholding methods will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In thresholding methods, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidencescore meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to thresholding methods. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so thresholding methods should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manualreview container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For thresholding methods, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to

Define measurable requirements for thresholding methods and record all assumptions.
Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for thresholding methods. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 17: MORPHOLOGICAL PROCESSING

This chapter expands the engineering treatment of morphological processing for the vision-based e-waste battery sorting line. The discussion is organised around erosion, dilation, opening, closing, structuring elements and defectregion refinement. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which morphological processing will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In morphological processing, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to morphological processing. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so morphological processing should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For morphological processing, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the

design. Calibration records and representative images are particularly valuable because they allow another engineer to Define measurable requirements for morphological processing and record all assumptions.

Check that the selected method is compatible with the conveyor speed and available processing time.

Test normal, damaged, reflective, partially occluded and low-quality examples.

Record per-class results and investigate every safety-relevant error.

Provide a reject or manual-review route for uncertain outcomes.

Store image IDs, timestamps, configuration and software/model versions.

Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for morphological processing. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 18: EDGE DETECTION

This chapter expands the engineering treatment of edge detection for the vision-based e-waste battery sorting line. The discussion is organised around Sobel gradients, Canny edge detection, edge continuity, parameter selection and false edges. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which edge detection will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In edge detection, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidencescore meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to edge detection. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so edge detection should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manualreview container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For edge detection, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design.

Calibration records and representative images are particularly valuable because they allow another engineer to Define measurable requirements for edge detection and record all assumptions.

Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for edge detection. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 19: SEGMENTATION STRATEGY

This chapter expands the engineering treatment of segmentation strategy for the vision-based e-waste battery sorting line. The discussion is organised around threshold-based segmentation, contour extraction, connected components and deep-learning segmentation. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which segmentation strategy will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In segmentation strategy, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidencescore meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to segmentation strategy. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so segmentation strategy should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manualreview container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For segmentation strategy, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to

Define measurable requirements for segmentation strategy and record all assumptions.
Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for segmentation strategy. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 20: OBJECT DETECTION STRATEGY

This chapter expands the engineering treatment of object detection strategy for the vision-based e-waste battery sorting line. The discussion is organised around bounding boxes, confidence scores, non-maximum suppression, class imbalance and localisation quality. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which object detection strategy will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In object detection strategy, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to object detection strategy. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so object detection strategy should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For object detection strategy, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the

design. Calibration records and representative images are particularly valuable because they allow another engineer to Define measurable requirements for object detection strategy and record all assumptions.

Check that the selected method is compatible with the conveyor speed and available processing time.

Test normal, damaged, reflective, partially occluded and low-quality examples.

Record per-class results and investigate every safety-relevant error.

Provide a reject or manual-review route for uncertain outcomes.

Store image IDs, timestamps, configuration and software/model versions.

Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for object detection strategy. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 21: OCR PIPELINE DESIGN

This chapter expands the engineering treatment of ocr pipeline design for the vision-based e-waste battery sorting line. The discussion is organised around text-region detection, perspective correction, character recognition, confidence and chemistry-string validation. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which ocr pipeline design will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In ocr pipeline design, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidencescore meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to ocr pipeline design. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so ocr pipeline design should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manualreview container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For ocr pipeline design, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design.

Calibration records and representative images are particularly valuable because they allow another engineer to Define measurable requirements for ocr pipeline design and record all assumptions.

Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for ocr pipeline design. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 22: OCR ERROR HANDLING

This chapter expands the engineering treatment of ocr error handling for the vision-based e-waste battery sorting line. The discussion is organised around confusable characters, damaged labels, low contrast, rotated text, partial labels and reject rules. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which ocr error handling will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In ocr error handling, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidencescore meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to ocr error handling. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so ocr error handling should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manualreview container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For ocr error handling, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design.

Calibration records and representative images are particularly valuable because they allow another engineer to Define measurable requirements for ocr error handling and record all assumptions.

Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for ocr error handling. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 23: DEFECT TAXONOMY

This chapter expands the engineering treatment of defect taxonomy for the vision-based e-waste battery sorting line. The discussion is organised around rust, dents, cracks, swelling, leakage indicators, contamination and surface abrasion. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which defect taxonomy will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In defect taxonomy, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidencescore meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to defect taxonomy. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so defect taxonomy should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manualreview container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For defect taxonomy, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design.

Calibration records and representative images are particularly valuable because they allow another engineer to Define measurable requirements for defect taxonomy and record all assumptions.

Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for defect taxonomy. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 24: DEFECT MEASUREMENT

This chapter expands the engineering treatment of defect measurement for the vision-based e-waste battery sorting line. The discussion is organised around area, length, width, contour irregularity, shape descriptors and calibrated physical measurements. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which defect measurement will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In defect measurement, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidencescore meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to defect measurement. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so defect measurement should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manualreview container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For defect measurement, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design.

Calibration records and representative images are particularly valuable because they allow another engineer to Define measurable requirements for defect measurement and record all assumptions.

Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for defect measurement. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 25: DATASET COLLECTION PLAN

This chapter expands the engineering treatment of dataset collection plan for the vision-based e-waste battery sorting line. The discussion is organised around representative sampling, operating shifts, battery suppliers, defect coverage and data governance. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which dataset collection plan will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In dataset collection plan, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidencescore meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to dataset collection plan. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so dataset collection plan should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manualreview container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For dataset collection plan, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to

Define measurable requirements for dataset collection plan and record all assumptions.
Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for dataset collection plan. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 26: DATASET ANNOTATION PROTOCOL

This chapter expands the engineering treatment of dataset annotation protocol for the vision-based e-waste battery sorting line. The discussion is organised around class definitions, bounding boxes, segmentation masks, OCR transcription and adjudication. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which dataset annotation protocol will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In dataset annotation protocol, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to dataset annotation protocol. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so dataset annotation protocol should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For dataset annotation protocol, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Engineering checklist

- Define measurable requirements for dataset annotation protocol and record all assumptions.
- Check that the selected method is compatible with the conveyor speed and available processing time.
- Test normal, damaged, reflective, partially occluded and low-quality examples.
- Record per-class results and investigate every safety-relevant error.
- Provide a reject or manual-review route for uncertain outcomes.
- Store image IDs, timestamps, configuration and software/model versions.
- Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for dataset annotation protocol. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 27: TRAINING AND VALIDATION SPLIT

This chapter expands the engineering treatment of training and validation split for the vision-based e-waste battery sorting line. The discussion is organised around train, validation and test separation, leakage prevention, batch identity and supplier diversity. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which training and validation split will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In training and validation split, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to training and validation split. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so training and validation split should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For training and validation split, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Engineering checklist

- Define measurable requirements for training and validation split and record all assumptions.
- Check that the selected method is compatible with the conveyor speed and available processing time.
- Test normal, damaged, reflective, partially occluded and low-quality examples.
- Record per-class results and investigate every safety-relevant error.
- Provide a reject or manual-review route for uncertain outcomes.
- Store image IDs, timestamps, configuration and software/model versions.
- Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for training and validation split. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 28: MODEL SELECTION RATIONALE

This chapter expands the engineering treatment of model selection rationale for the vision-based e-waste battery sorting line. The discussion is organised around classical computer vision, convolutional networks, object detectors, OCR engines and hybrid architectures. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which model selection rationale will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In model selection rationale, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to model selection rationale. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so model selection rationale should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For model selection rationale, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was

needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Define measurable requirements for model selection rationale and record all assumptions.
Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for model selection rationale. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 29: MODEL TRAINING WORKFLOW

This chapter expands the engineering treatment of model training workflow for the vision-based e-waste battery sorting line. The discussion is organised around pre-training, augmentation, optimisation, learning rate, epochs, early stopping and reproducibility. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which model training workflow will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In model training workflow, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to model training workflow. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so model training workflow should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For model training workflow, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Engineering checklist

- Define measurable requirements for model training workflow and record all assumptions.
- Check that the selected method is compatible with the conveyor speed and available processing time.
- Test normal, damaged, reflective, partially occluded and low-quality examples.
- Record per-class results and investigate every safety-relevant error.
- Provide a reject or manual-review route for uncertain outcomes.
- Store image IDs, timestamps, configuration and software/model versions.
- Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for model training workflow. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 30: DATA AUGMENTATION STRATEGY

This chapter expands the engineering treatment of data augmentation strategy for the vision-based e-waste battery sorting line. The discussion is organised around rotation, translation, scale, brightness, contrast, blur, occlusion and physically realistic transformations. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which data augmentation strategy will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In data augmentation strategy, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to data augmentation strategy. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so data augmentation strategy should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For data augmentation strategy, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Engineering checklist

- Define measurable requirements for data augmentation strategy and record all assumptions.
- Check that the selected method is compatible with the conveyor speed and available processing time.
- Test normal, damaged, reflective, partially occluded and low-quality examples.
- Record per-class results and investigate every safety-relevant error.
- Provide a reject or manual-review route for uncertain outcomes.
- Store image IDs, timestamps, configuration and software/model versions.
- Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for data augmentation strategy. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 31: CLASS IMBALANCE MANAGEMENT

This chapter expands the engineering treatment of class imbalance management for the vision-based e-waste battery sorting line. The discussion is organised around weighted loss, oversampling, targeted data collection and per-class evaluation. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which class imbalance management will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In class imbalance management, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to class imbalance management. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so class imbalance management should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For class imbalance management, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Engineering checklist

- Define measurable requirements for class imbalance management and record all assumptions.
- Check that the selected method is compatible with the conveyor speed and available processing time.
- Test normal, damaged, reflective, partially occluded and low-quality examples.
- Record per-class results and investigate every safety-relevant error.
- Provide a reject or manual-review route for uncertain outcomes.
- Store image IDs, timestamps, configuration and software/model versions.
- Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for class imbalance management. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 32: PERFORMANCE METRICS

This chapter expands the engineering treatment of performance metrics for the vision-based e-waste battery sorting line. The discussion is organised around accuracy, precision, recall, F1-score, confusion matrix, IoU and mean average precision. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which performance metrics will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In performance metrics, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidencescore meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to performance metrics. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so performance metrics should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manualreview container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For performance metrics, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Engineering checklist

- Define measurable requirements for performance metrics and record all assumptions.
- Check that the selected method is compatible with the conveyor speed and available processing time.
- Test normal, damaged, reflective, partially occluded and low-quality examples.
- Record per-class results and investigate every safety-relevant error.
- Provide a reject or manual-review route for uncertain outcomes.
- Store image IDs, timestamps, configuration and software/model versions.
- Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for performance metrics. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 33: SAFETY-ORIENTED DECISION THRESHOLDS

This chapter expands the engineering treatment of safety-oriented decision thresholds for the vision-based e-waste battery sorting line. The discussion is organised around false acceptance, false rejection, confidence calibration, reject bias and human review. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which safety-oriented decision thresholds will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In safety-oriented decision thresholds, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to safetyoriented decision thresholds. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so safety-oriented decision thresholds should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For safety-oriented decision thresholds, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change,

why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Engineering checklist

- Define measurable requirements for safety-oriented decision thresholds and record all assumptions.
- Check that the selected method is compatible with the conveyor speed and available processing time.
- Test normal, damaged, reflective, partially occluded and low-quality examples.
- Record per-class results and investigate every safety-relevant error.
- Provide a reject or manual-review route for uncertain outcomes.
- Store image IDs, timestamps, configuration and software/model versions.
- Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for safety-oriented decision thresholds. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 34: REAL-TIME PROCESSING ARCHITECTURE

This chapter expands the engineering treatment of real-time processing architecture for the vision-based e-waste battery sorting line. The discussion is organised around frame acquisition, buffering, inference, decision logic, PLC communication and actuator timing. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which real-time processing architecture will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In real-time processing architecture, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to real-time processing architecture. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so real-time processing architecture should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For real-time processing architecture, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Engineering checklist

- Define measurable requirements for real-time processing architecture and record all assumptions.
- Check that the selected method is compatible with the conveyor speed and available processing time.

- Test normal, damaged, reflective, partially occluded and low-quality examples.
- Record per-class results and investigate every safety-relevant error.
- Provide a reject or manual-review route for uncertain outcomes.
- Store image IDs, timestamps, configuration and software/model versions.
- Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for real-time processing architecture. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 35: CONVEYOR TRACKING AND SYNCHRONISATION

This chapter expands the engineering treatment of conveyor tracking and synchronisation for the vision-based e-waste battery sorting line. The discussion is organised around encoder pulses, object IDs, tracking windows, trigger offsets and reject-station coordination. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which conveyor tracking and synchronisation will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In conveyor tracking and synchronisation, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to conveyor tracking and synchronisation. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so conveyor tracking and synchronisation should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For conveyor tracking and synchronisation, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Engineering checklist

- Define measurable requirements for conveyor tracking and synchronisation and record all assumptions.
- Check that the selected method is compatible with the conveyor speed and available processing time.
- Test normal, damaged, reflective, partially occluded and low-quality examples.
- Record per-class results and investigate every safety-relevant error.
- Provide a reject or manual-review route for uncertain outcomes.
- Store image IDs, timestamps, configuration and software/model versions.
- Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for conveyor tracking and synchronisation. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 36: PNEUMATIC REJECTER DESIGN

This chapter expands the engineering treatment of pneumatic rejecter design for the vision-based e-waste battery sorting line. The discussion is organised around valve response, cylinder stroke, air pressure, pulse duration, object spacing and safe rejection. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which pneumatic rejecter design will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In pneumatic rejecter design, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to pneumatic rejecter design. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so pneumatic rejecter design should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For pneumatic rejecter design, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the

design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Define measurable requirements for pneumatic rejecter design and record all assumptions.
Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for pneumatic rejecter design. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 37: PLC AND INDUSTRIAL COMMUNICATION

This chapter expands the engineering treatment of plc and industrial communication for the vision-based e-waste battery sorting line. The discussion is organised around digital I/O, Ethernet protocols, handshaking, watchdogs, interlocks and fault states. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which plc and industrial communication will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In plc and industrial communication, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to plc and industrial communication. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so plc and industrial communication should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For plc and industrial communication, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Define measurable requirements for plc and industrial communication and record all assumptions.
Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for plc and industrial communication. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 38: SYSTEM STATE MACHINE

This chapter expands the engineering treatment of system state machine for the vision-based e-waste battery sorting line. The discussion is organised around idle, acquire, inspect, classify, track, reject, confirm, fault and recovery states. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which system state machine will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In system state machine, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidencescore meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to system state machine. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so system state machine should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manualreview container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For system state machine, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design.

Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Define measurable requirements for system state machine and record all assumptions.
Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for system state machine. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 39: FAIL-SAFE OPERATION

This chapter expands the engineering treatment of fail-safe operation for the vision-based e-waste battery sorting line. The discussion is organised around unknown chemistry, missing image, low confidence, sensor failure, communication loss and emergency stop. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which fail-safe operation will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In fail-safe operation, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidencescore meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to fail-safe operation. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so fail-safe operation should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manualreview container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For fail-safe operation, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Define measurable requirements for fail-safe operation and record all assumptions.
Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for fail-safe operation. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 40: HUMAN-MACHINE INTERFACE

This chapter expands the engineering treatment of human-machine interface for the vision-based e-waste battery sorting line. The discussion is organised around operator dashboard, live image, class result, confidence, alarms, audit trail and access control. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which human-machine interface will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In human-machine interface, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to humanmachine interface. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so human-machine interface should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For human-machine interface, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the

design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Define measurable requirements for human-machine interface and record all assumptions.
Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for human-machine interface. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 41: TRACEABILITY AND DATA LOGGING

This chapter expands the engineering treatment of traceability and data logging for the vision-based e-waste battery sorting line. The discussion is organised around timestamps, image IDs, model version, decisions, operator overrides and retention policy. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which traceability and data logging will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In traceability and data logging, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to traceability and data logging. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so traceability and data logging should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For traceability and data logging, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Define measurable requirements for traceability and data logging and record all assumptions.
Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for traceability and data logging. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 42: CYBERSECURITY AND ACCESS CONTROL

This chapter expands the engineering treatment of cybersecurity and access control for the vision-based e-waste battery sorting line. The discussion is organised around network segmentation, least privilege, secure updates, credentials, backups and audit logs. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which cybersecurity and access control will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In cybersecurity and access control, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to cybersecurity and access control. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so cybersecurity and access control should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For cybersecurity and access control, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Define measurable requirements for cybersecurity and access control and record all assumptions.
Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for cybersecurity and access control. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 43: RELIABILITY AND MAINTENANCE

This chapter expands the engineering treatment of reliability and maintenance for the vision-based e-waste battery sorting line. The discussion is organised around cleaning, lens inspection, lighting checks, calibration checks, spare parts and preventive maintenance. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which reliability and maintenance will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In reliability and maintenance, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to reliability and maintenance. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so reliability and maintenance should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For reliability and maintenance, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the

design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Define measurable requirements for reliability and maintenance and record all assumptions.
Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for reliability and maintenance. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 44: FACTORY ACCEPTANCE TESTING

This chapter expands the engineering treatment of factory acceptance testing for the vision-based e-waste battery sorting line. The discussion is organised around test cases, throughput trials, defect challenge samples, safety checks and acceptance evidence. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which factory acceptance testing will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In factory acceptance testing, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to factory acceptance testing. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so factory acceptance testing should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For factory acceptance testing, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the

design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Define measurable requirements for factory acceptance testing and record all assumptions.
Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for factory acceptance testing. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 45: SITE ACCEPTANCE TESTING

This chapter expands the engineering treatment of site acceptance testing for the vision-based e-waste battery sorting line. The discussion is organised around installation checks, production trials, operator training, baseline comparison and sign-off. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which site acceptance testing will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In site acceptance testing, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidencescore meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to site acceptance testing. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so site acceptance testing should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station,

the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manualreview container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For site acceptance testing, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to

Define measurable requirements for site acceptance testing and record all assumptions.
Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for site acceptance testing. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 46: RISK ASSESSMENT AND FMEA

This chapter expands the engineering treatment of risk assessment and fmea for the vision-based e-waste battery sorting line. The discussion is organised around failure modes, causes, effects, severity, occurrence, detection and mitigation. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which risk assessment and fmea will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In risk assessment and fmea, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to risk assessment and fmea. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so risk assessment and fmea should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject

station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For risk assessment and fmea, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to Define measurable requirements for risk assessment and fmea and record all assumptions.

Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for risk assessment and fmea. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 47: ENVIRONMENTAL ROBUSTNESS

This chapter expands the engineering treatment of environmental robustness for the vision-based e-waste battery sorting line. The discussion is organised around dust, vibration, temperature, humidity, ambient light and enclosure protection. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which environmental robustness will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In environmental robustness, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to environmental robustness. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so environmental robustness should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject

station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For environmental robustness, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to Define measurable requirements for environmental robustness and record all assumptions.

Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for environmental robustness. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 48: ENERGY AND OPERATIONAL EFFICIENCY

This chapter expands the engineering treatment of energy and operational efficiency for the vision-based e-waste battery sorting line. The discussion is organised around camera and compute power, compressed air use, reject rate, downtime and resource efficiency. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which energy and operational efficiency will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In energy and operational efficiency, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to energy and operational efficiency. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so energy and operational efficiency should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For energy and operational efficiency, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Define measurable requirements for energy and operational efficiency and record all assumptions.
Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for energy and operational efficiency. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 49: COST-BENEFIT ANALYSIS

This chapter expands the engineering treatment of cost-benefit analysis for the vision-based e-waste battery sorting line. The discussion is organised around capital cost, integration cost, labour impact, scrap prevention, downtime and payback assumptions. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which cost-benefit analysis will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In cost-benefit analysis, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidencescore meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to costbenefit analysis. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so cost-benefit analysis should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manualreview container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For cost-benefit analysis, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design.

Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Define measurable requirements for cost-benefit analysis and record all assumptions.
Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for cost-benefit analysis. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 50: SCALABILITY AND FUTURE IMPROVEMENTS

This chapter expands the engineering treatment of scalability and future improvements for the vision-based e-waste battery sorting line. The discussion is organised around multiple lanes, additional battery formats, model updates, edge computing and modular expansion. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which scalability and future improvements will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In scalability and future improvements, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to scalability and future improvements. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so scalability and future improvements should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For scalability and future improvements, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Define measurable requirements for scalability and future improvements and record all assumptions.
Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for scalability and future improvements. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 51: ETHICAL AND DATA CONSIDERATIONS

This chapter expands the engineering treatment of ethical and data considerations for the vision-based e-waste battery sorting line. The discussion is organised around worker privacy, image scope, retention, transparent monitoring and responsible automation. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which ethical and data considerations will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In ethical and data considerations, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to ethical and data considerations. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so ethical and data considerations should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For ethical and data considerations, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the

design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Define measurable requirements for ethical and data considerations and record all assumptions.
Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for ethical and data considerations. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 52: PROJECT IMPLEMENTATION SCHEDULE

This chapter expands the engineering treatment of project implementation schedule for the vision-based e-waste battery sorting line. The discussion is organised around requirements, procurement, installation, data collection, development, validation and commissioning. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which project implementation schedule will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In project implementation schedule, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to project implementation schedule. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so project implementation schedule should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For project implementation schedule, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Define measurable requirements for project implementation schedule and record all assumptions.
Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for project implementation schedule. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 53: TECHNICAL DOCUMENTATION STRUCTURE

This chapter expands the engineering treatment of technical documentation structure for the vision-based e-waste battery sorting line. The discussion is organised around system drawings, wiring, calibration records, model cards, test reports and operating procedures. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which technical documentation structure will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In technical documentation structure, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to technical documentation structure. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so technical documentation structure should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For technical documentation structure, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Define measurable requirements for technical documentation structure and record all assumptions.
Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for technical documentation structure. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 54: TROUBLESHOOTING GUIDE

This chapter expands the engineering treatment of troubleshooting guide for the vision-based e-waste battery sorting line. The discussion is organised around blur, glare, OCR failure, false defect, missed object, delayed reject and communication alarms. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which troubleshooting guide will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In troubleshooting guide, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidencescore meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to troubleshooting guide. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so troubleshooting guide should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manualreview container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For troubleshooting guide, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design.

Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Define measurable requirements for troubleshooting guide and record all assumptions.
Check that the selected method is compatible with the conveyor speed and available processing time.
Test normal, damaged, reflective, partially occluded and low-quality examples.
Record per-class results and investigate every safety-relevant error.
Provide a reject or manual-review route for uncertain outcomes.
Store image IDs, timestamps, configuration and software/model versions.
Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for troubleshooting guide. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 55: OPERATOR STANDARD OPERATING PROCEDURE

This chapter expands the engineering treatment of operator standard operating procedure for the vision-based e-waste battery sorting line. The discussion is organised around start-up, sample checks, monitoring, exception handling, shutdown and escalation. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which operator standard operating procedure will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In operator standard operating procedure, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to operator standard operating procedure. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so operator standard operating procedure should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor

should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For operator standard operating procedure, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change,

why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Engineering checklist

- Define measurable requirements for operator standard operating procedure and record all assumptions.
- Check that the selected method is compatible with the conveyor speed and available processing time.
- Test normal, damaged, reflective, partially occluded and low-quality examples.
- Record per-class results and investigate every safety-relevant error.
- Provide a reject or manual-review route for uncertain outcomes.
- Store image IDs, timestamps, configuration and software/model versions.
- Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for operator standard operating procedure. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 56: QUALITY ASSURANCE PLAN

This chapter expands the engineering treatment of quality assurance plan for the vision-based e-waste battery sorting line. The discussion is organised around inspection sampling, golden samples, drift checks, version control and periodic audits. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which quality assurance plan will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In quality assurance plan, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidencescore meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to quality assurance plan. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so quality assurance plan should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manualreview container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For quality assurance plan, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Engineering checklist

- Define measurable requirements for quality assurance plan and record all assumptions.
- Check that the selected method is compatible with the conveyor speed and available processing time.

- Test normal, damaged, reflective, partially occluded and low-quality examples.
- Record per-class results and investigate every safety-relevant error.
- Provide a reject or manual-review route for uncertain outcomes.
- Store image IDs, timestamps, configuration and software/model versions.
- Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for quality assurance plan. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 57: MODEL DRIFT MONITORING

This chapter expands the engineering treatment of model drift monitoring for the vision-based e-waste battery sorting line. The discussion is organised around supplier changes, print changes, lighting changes, distribution shift and retraining triggers. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which model drift monitoring will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In model drift monitoring, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidencescore meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to model drift monitoring. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so model drift monitoring should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For model drift monitoring, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Engineering checklist

- Define measurable requirements for model drift monitoring and record all assumptions.
- Check that the selected method is compatible with the conveyor speed and available processing time.

- Test normal, damaged, reflective, partially occluded and low-quality examples.
- Record per-class results and investigate every safety-relevant error.
- Provide a reject or manual-review route for uncertain outcomes.
- Store image IDs, timestamps, configuration and software/model versions.
- Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for model drift monitoring. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 58: COMPARATIVE EVALUATION OF METHODS

This chapter expands the engineering treatment of comparative evaluation of methods for the vision-based e-waste battery sorting line. The discussion is organised around manual inspection, assisted inspection and automated vision using common criteria. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which comparative evaluation of methods will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In comparative evaluation of methods, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to comparative evaluation of methods. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so comparative evaluation of methods should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For comparative evaluation of methods, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Engineering checklist

- Define measurable requirements for comparative evaluation of methods and record all assumptions.
- Check that the selected method is compatible with the conveyor speed and available processing time.
- Test normal, damaged, reflective, partially occluded and low-quality examples.
- Record per-class results and investigate every safety-relevant error.
- Provide a reject or manual-review route for uncertain outcomes.
- Store image IDs, timestamps, configuration and software/model versions.
- Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for comparative evaluation of methods. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 59: OVERALL SYSTEM INTEGRATION REVIEW

This chapter expands the engineering treatment of overall system integration review for the vision-based e-waste battery sorting line. The discussion is organised around mechanical, optical, computational, electrical, software and safety integration. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which overall system integration review will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In overall system integration review, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to overall system integration review. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so overall system integration review should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For overall system integration review, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Engineering checklist

- Define measurable requirements for overall system integration review and record all assumptions.

- Check that the selected method is compatible with the conveyor speed and available processing time.
- Test normal, damaged, reflective, partially occluded and low-quality examples.
- Record per-class results and investigate every safety-relevant error.
- Provide a reject or manual-review route for uncertain outcomes.
- Store image IDs, timestamps, configuration and software/model versions.
- Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for overall system integration review. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

SUPPLEMENTARY TECHNICAL CHAPTER 60: CONCLUSION AND ENGINEERING RECOMMENDATIONS

This chapter expands the engineering treatment of conclusion and engineering recommendations for the vision-based e-waste battery sorting line. The discussion is organised around design priorities, commissioning evidence, operational controls and future development. The purpose is to explain the design choices, identify the variables that influence performance, and show how the subject should be verified during implementation. These details complement the task answers in the main assignment; they do not represent measurements from a completed factory installation.

The first consideration is to define the operating condition in which conclusion and engineering recommendations will be used. The system receives mixed batteries on a moving conveyor, so the solution must tolerate differences in size, orientation, surface finish, printing quality and environmental conditions. A method that works on a clean laboratory image may fail when the same object is dusty, scratched, partially hidden or moving. Therefore, design inputs should be recorded before selecting algorithms or hardware. Important inputs include the smallest object feature that must be detected, the maximum conveyor speed, the expected distance between adjacent cells, the range of ambient illumination, and the acceptable false-accept and false-reject rates. Each input should have a stated unit, a source and a verification method. This makes later calculations auditable and prevents assumptions from being mistaken for measured facts.

A practical implementation should separate the process into acquisition, image conditioning, feature extraction, interpretation and decision stages. In conclusion and engineering recommendations, the quality of an early stage places a limit on every stage that follows. For example, poor focus cannot reliably be repaired by a more complex classifier, and clipped highlights can remove label details that OCR needs. Similarly, an accurate defect detector cannot compensate for a conveyor trigger that associates the wrong image with the wrong physical battery. The engineering team should therefore define interfaces between stages: image dimensions and pixel format, coordinate conventions, object identifiers, confidence-score meaning, timestamps and error codes. Intermediate outputs should be logged during commissioning so that faults can be traced to the stage where they originate.

Verification should use a controlled test set containing normal examples and difficult edge cases relevant to conclusion and engineering recommendations. Samples should cover each supported form factor, different suppliers and label conditions, multiple orientations, reflective and damaged surfaces, and representative defect severities. The team should record both successful and unsuccessful cases rather than reporting only an overall accuracy value. For a classification stage, a confusion matrix shows which classes are being confused. For localisation, intersection over union can describe overlap between predicted and reference regions. For OCR, character error rate and exact chemistry-string accuracy are useful. For the complete line, end-to-end throughput, missed-reject rate and false-reject rate must also be measured. Test conditions, software version, model version and sample identifiers should accompany every result.

Safety and uncertainty must be considered explicitly. The sorting line handles battery types that may require different downstream handling, so conclusion and engineering recommendations should not be treated as a cosmetic feature. If an image is missing, the label is unreadable, a defect score is close to its decision threshold, or the object cannot be tracked to the reject station, the system should move to a defined safe state. A conservative design can route uncertain cells to a reject or manual-review container. This reduces the chance that an uncertain object is silently accepted. The reject path itself must be verified: the PLC command, pneumatic response, physical ejection and confirmation sensor should all be checked. An alarm should be generated when a command is not confirmed within its allowed time window.

Documentation is part of the technical solution. For conclusion and engineering recommendations, the report should state the chosen method, alternatives considered, parameter values, assumptions, known limitations and evidence used to accept the design. Calibration records and representative images are particularly valuable because they allow another engineer to reproduce the setup. Changes to the camera, lens, lighting, conveyor speed or model may affect performance and should trigger an impact assessment. A controlled change log should identify who made the change, why it was needed, which version was installed and which tests were repeated. This practice supports maintenance, troubleshooting and long-term reliability.

Engineering checklist

- Define measurable requirements for conclusion and engineering recommendations and record all assumptions.
- Check that the selected method is compatible with the conveyor speed and available processing time.
- Test normal, damaged, reflective, partially occluded and low-quality examples.
- Record per-class results and investigate every safety-relevant error.
- Provide a reject or manual-review route for uncertain outcomes.
- Store image IDs, timestamps, configuration and software/model versions.
- Repeat verification after any relevant hardware, lighting or software change.

Expected engineering output

The expected output of this chapter is a documented and testable design for conclusion and engineering recommendations. The design should identify its inputs, processing steps, output format, operating limits, failure response and verification evidence. During actual deployment, proposed targets must be replaced or confirmed using measurements from the installed system and a labelled production-representative dataset. No performance value should be described as experimentally achieved unless the corresponding test was performed and recorded.

APPENDIX A – TECHNICAL GLOSSARY

Acquisition: The process of capturing a digital image using a camera and sensor.

Aperture: The opening in a lens that controls the amount of light reaching the sensor.

Bounding box: A rectangular region used to indicate the location of an object or defect.

Calibration: The process of determining camera or system parameters using known references.

Canny edge detector: A multi-stage edge detector designed to identify strong and continuous image boundaries.

CLAHE: Contrast Limited Adaptive Histogram Equalisation, a local contrast enhancement method.

Confidence score: A model output indicating its estimated support for a prediction; it is not automatically a calibrated probability.

Connected component: A set of foreground pixels connected under a chosen neighbourhood rule.

Depth of field: The range of object distances that appear acceptably sharp in an image.

Exposure time: The time interval during which the sensor collects light for one image.

False acceptance: An unsafe or incorrect object is incorrectly accepted by the system.

False rejection: A suitable object is incorrectly rejected by the system.

Field of view: The physical scene area visible through the camera and lens.

Focal length: An optical parameter that influences magnification and field of view.

Frame rate: The number of images acquired or processed per second.

Histogram: A count of pixel values or colour-channel values in an image.

IoU: Intersection over Union, a measure of overlap between two regions.

OCR: Optical Character Recognition, the conversion of image text into machine-readable characters.

Pixel pitch: The centre-to-centre spacing between adjacent sensor pixels.

Quantisation: Mapping continuous or high-precision values into a finite set of digital levels.

Region of interest: A selected image area used for focused processing.

Recall: The fraction of actual positive cases correctly detected.

Precision: The fraction of predicted positive cases that are actually positive.

Threshold: A decision boundary used to convert a score or pixel value into a category.

Trigger: A signal that starts image acquisition or synchronises an operation.

White balance: Adjustment of colour-channel gains to compensate for illumination colour.

APPENDIX B – FORMULA AND METRIC REFERENCE

Cell interval: $T = 1 / R$, where R is the arrival rate in cells per second.

Required pixels per millimetre: P = image pixels covering the object / object width in millimetres.

Object-space pixel size: s = field-of-view width / image width in pixels.

Motion blur in object space: b = conveyor velocity × exposure time.

Frames required per second: $F \geq$ object arrival rate × required frames per object.

Precision: $TP / (TP + FP)$.

Recall: $TP / (TP + FN)$.

F1-score: $2 \times \text{precision} \times \text{recall} / (\text{precision} + \text{recall})$.

Intersection over Union: Area of intersection / area of union.

Character error rate: (substitutions + deletions + insertions) / number of reference characters.

Throughput: Number of correctly processed objects / elapsed time.

Reject rate: Number of rejected objects / total inspected objects.

The formulas in this appendix are reference definitions. Their use requires consistent units and clearly defined counting rules. For example, throughput should state whether it counts all inspected cells or only correctly classified cells. Precision and recall should be reported for each class when class frequencies differ. A single aggregate score can hide poor performance on a rare but safety-critical battery type.

APPENDIX C – FINAL COMMISSIONING RECORD TEMPLATE

Project / line identification: _____

Camera make, model and serial number: _____

Lens and focal length: _____

Lighting type and configuration: _____

Conveyor speed and measurement method: _____

Image resolution and frame rate: _____

Calibration date and reference target: _____

Model name, version and checksum: _____

Dataset version and test-set size: _____

Measured throughput: _____

Per-class precision / recall / F1: _____

OCR exact-string accuracy: _____

Defect detection results: _____

Reject actuator response time: _____

Missed-reject test result: _____

Outstanding issues and corrective actions: _____

Tested by / date: _____

Approved by / date: _____